

Rectilinear Kinematics (Ch. 12)

$v = \frac{ds}{dt}$ – velocity is the time rate of change of position

$a = \frac{dv}{dt}$ – acceleration is the time rate of change of velocity

$a ds = v dv$ – eliminates time; use when a is a function of position

$v = v_0 + a_c t$

$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$

$v^2 = v_0^2 + 2a_c(s - s_0)$ – constant a_c only; keep sign convention consistent

s - t , v - t , a - t graphs:

v = slope of the s - t graph at that instant (ds/dt)

a = slope of the v - t graph at that instant (dv/dt)

Δv = area under the a - t graph (change in velocity)

Δs = area under the v - t graph (displacement)

Variable acceleration – which relation to use:

$a = f(t)$: integrate directly

$a = f(v)$: use $a = dv/dt$, separate variables

$a = f(s)$: use $a ds = v dv$, separate variables

Curvilinear & Projectile Motion (Ch. 12)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$ – $\mathbf{v}$ is always tangent to the path

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{\mathbf{r}}$ – $\mathbf{a}$ is generally NOT tangent to the path in curvilinear motion

$v_x = \dot{x}$, $v_y = \dot{y}$, $v_z = \dot{z}$ – rectangular velocity components; each is an independent rectilinear problem

$a_x = \ddot{x}$, $a_y = \ddot{y}$, $a_z = \ddot{z}$ – rectangular acceleration components; combine vectorially

$x = x_0 + (v_x)_0 t$

$y = y_0 + (v_y)_0 t - \frac{1}{2} g t^2$ – $a_x = 0$ (constant v_x), $a_y = -g$ ($g \approx 9.81 \text{ m/s}^2$)

$v_y = (v_y)_0 - g t$

$v_y^2 = (v_y)_0^2 - 2g(y - y_0)$ – solve x - and y -motion separately, linked by common time t

$v = \sqrt{v_x^2 + v_y^2}$ – resultant speed

$\theta = \tan^{-1}(v_y/v_x)$ – measured from x -axis to velocity vector; same approach for acceleration vector

Normal-Tangential (n-t) Coordinates (Ch. 12)

$v = \dot{s} \hat{u}_t$ – velocity always tangent to path, direction of increasing s

$\mathbf{a} = a_t \hat{u}_t + a_n \hat{u}_n$ – acceleration splits into tangential and normal parts

$a_t = \dot{v} = v \frac{dv}{ds}$ – tangential component; changes the *speed*

$a_n = \frac{v^2}{\rho}$ – normal component; changes *direction*; always points toward the center of curvature

$a = \sqrt{a_t^2 + a_n^2}$ – resultant magnitude of $\mathbf{a}$

$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$ – radius of curvature, needed for a_n when path $y = f(x)$ is analytic

Circular path: $\rho = \text{const} = r$

Straight-line path: $\rho \rightarrow \infty \Rightarrow a_n = 0$, $a = a_t$ – n - t coordinates reduce to ordinary rectilinear motion

Polar & Cylindrical Coordinates (Ch. 12)

$\mathbf{r} = r \hat{u}_r$ – position

$v_r = \dot{r}$ – radial velocity component

$v_\theta = r \dot{\theta}$ – transverse velocity component

$v = \sqrt{v_r^2 + v_\theta^2}$ – always tangent to the path, as it must be

$a_r = \ddot{r} - r \dot{\theta}^2$ – radial acceleration component

$a_\theta = r \ddot{\theta} + 2\dot{r}\dot{\theta}$ – transverse; the $2\dot{r}\dot{\theta}$ term is essential whenever r varies with time

$a = \sqrt{a_r^2 + a_\theta^2}$ – resultant magnitude

$\dot{\hat{u}}_r = \dot{\theta} \hat{u}_\theta$, $\dot{\hat{u}}_\theta = -\dot{\theta} \hat{u}_r$ – unit-vector rates; source of the extra terms above

$v_z = \dot{z}$, $a_z = \ddot{z}$ – cylindrical adds the z -axis; combine vectorially:

$\mathbf{v} = v_r \hat{u}_r + v_\theta \hat{u}_\theta + v_z \hat{u}_z$

Circular path in polar form ($r = \text{const}$): $a_r = -r \dot{\theta}^2$, $a_\theta = r \ddot{\theta}$ – matches n - t with $\rho = r$

Dependent & Relative Motion (Ch. 12)

$s_A + l_{CD} + s_B = l_T$ – write a position-coordinate equation using the total constant cable length

$\dot{s}_A + \dot{s}_B = 0$ – differentiate once for velocities

$\ddot{s}_A + \ddot{s}_B = 0$ – differentiate again for accelerations; opposite signs mean the blocks move oppositely (coefficients = number of cable segments changing length at each moving pulley)

$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$ – relative position; translating, non-rotating $x'y'$ frame attached to A

$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$ – relative velocity

$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$ – relative acceleration; for two independently-moving particles (e.g. two aircraft, a boat crossing a current) – solve as a vector equation in components

Kinetics: Equations of Motion (Ch. 13)

$\Sigma \mathbf{F} = m\mathbf{a}$ – draw a free-body diagram AND a kinetic ($m\mathbf{a}$) diagram before writing scalar equations

Rectangular components – best when path/forces are naturally in x, y, z (incline, drag/thrust problems):

$\Sigma F_x = ma_x$, $\Sigma F_y = ma_y$

$\Sigma F_z = ma_z$

Normal-tangential components – use when ρ is known (circular motion, banked curves, vertical loops):

$\Sigma F_t = ma_t = m \frac{dv}{dt}$

$\Sigma F_n = ma_n = m \frac{v^2}{\rho}$

Polar/cylindrical components – use when r, θ are the natural coordinates (e.g. rotating slotted arm):

$\Sigma F_r = m(\ddot{r} - r \dot{\theta}^2)$

$\Sigma F_\theta = m(r \ddot{\theta} + 2\dot{r}\dot{\theta})$

$\Sigma F_z = m\ddot{z}$

$\frac{d^2u}{d\theta^2} + u = -\frac{F}{mh^2u^2}$, $u = \frac{1}{r}$ – central-force/orbital motion; $F > 0$ away from force center O

$h = r^2 \dot{\theta} = \text{const}$ – angular momentum per unit mass, constant for any central force

Kinetics: Work & Energy (Ch. 14)

$U_{1-2} = \int_{s_1}^{s_2} F \cos \theta ds$ – positive when F has a component along motion

$U_W = -W\Delta y - \Delta y$ positive upward

$U_s = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right) - s$ from unstretched length; negative when stretching

$T_1 + \Sigma U_{1-2} = T_2$ - principle of work and energy (scalar); include friction as negative work

$T = \frac{1}{2}mv^2$ - kinetic energy of a particle

$P = \frac{dU}{dt} = \mathbf{F} \cdot \mathbf{v}$ - power, in watts (J/s)

$\varepsilon = \frac{P_{out}}{P_{in}} = \frac{\text{energy output}}{\text{energy input}}$ - efficiency; always < 1 (friction, heat losses)

$V_g = Wy$ - gravitational potential energy, measured from a chosen datum (+ above, - below)

$V_e = \frac{1}{2}ks^2$ - elastic potential energy; always ≥ 0 (s = deformation from unstretched length)

$T_1 + V_1 = T_2 + V_2$ - conservation of mechanical energy; valid only if all forces doing work are conservative

Kinetics: Impulse & Momentum (*Ch. 15*)

$m\mathbf{v}_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$ - linear impulse-momentum; useful when $F = f(t)$

$\Sigma m_A(v_A)_1 + \Sigma m_B(v_B)_1 = \Sigma m_A(v_A)_2 + \Sigma m_B(v_B)_2$ - conservation, e.g. collisions

$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1} - e = 1$ perfectly elastic (KE conserved), $e = 0$

perfectly plastic (move together)

Oblique impact: resolve along line of impact (n) and plane of contact (t); momentum & e apply along n , tangential v unchanged (smooth surfaces)

$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v}$ - angular momentum of a particle about O

$H_O = mvd - d =$ perpendicular distance from O to the line of action of $m\mathbf{v}$

$H_O = mr^2\dot{\theta} = mrv_\theta$ - angular momentum in polar coordinates

$(H_O)_1 + \Sigma \int_{t_1}^{t_2} M_O dt = (H_O)_2$ - angular impulse-momentum; moments summed about the same fixed O

H_O is conserved (constant) whenever $\Sigma M_O = 0$, as in central-force motion

Quick Reference

$\Sigma \mathbf{F} = m\mathbf{a}$ - Newton's second law; master equation for every kinetics problem

$v = v_0 + a_ct$, $s = s_0 + v_0t + \frac{1}{2}a_ct^2$, $v^2 = v_0^2 + 2a_c(s - s_0)$ - constant acceleration

$a_t = v dv/ds$, $a_n = v^2/\rho$ - n - t acceleration components

$a_r = \ddot{r} - r\dot{\theta}^2$, $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ - polar acceleration components

$T_1 + \Sigma U_{1-2} = T_2$ - work-energy principle

$m\mathbf{v}_1 + \Sigma \int \mathbf{F} dt = m\mathbf{v}_2$ - linear impulse-momentum

$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$ - coefficient of restitution